GROUP 5 · IIT MADRAS DS AND AI LAB · 2026 T3 · MILESTONE 2

SerpentGuard AI — Dataset Preparation & EDA

Offline Indian Snake Segmentation, Classification & Safety Decision Support (`puneetbajajiitm/serpentguard-m2-splits`)

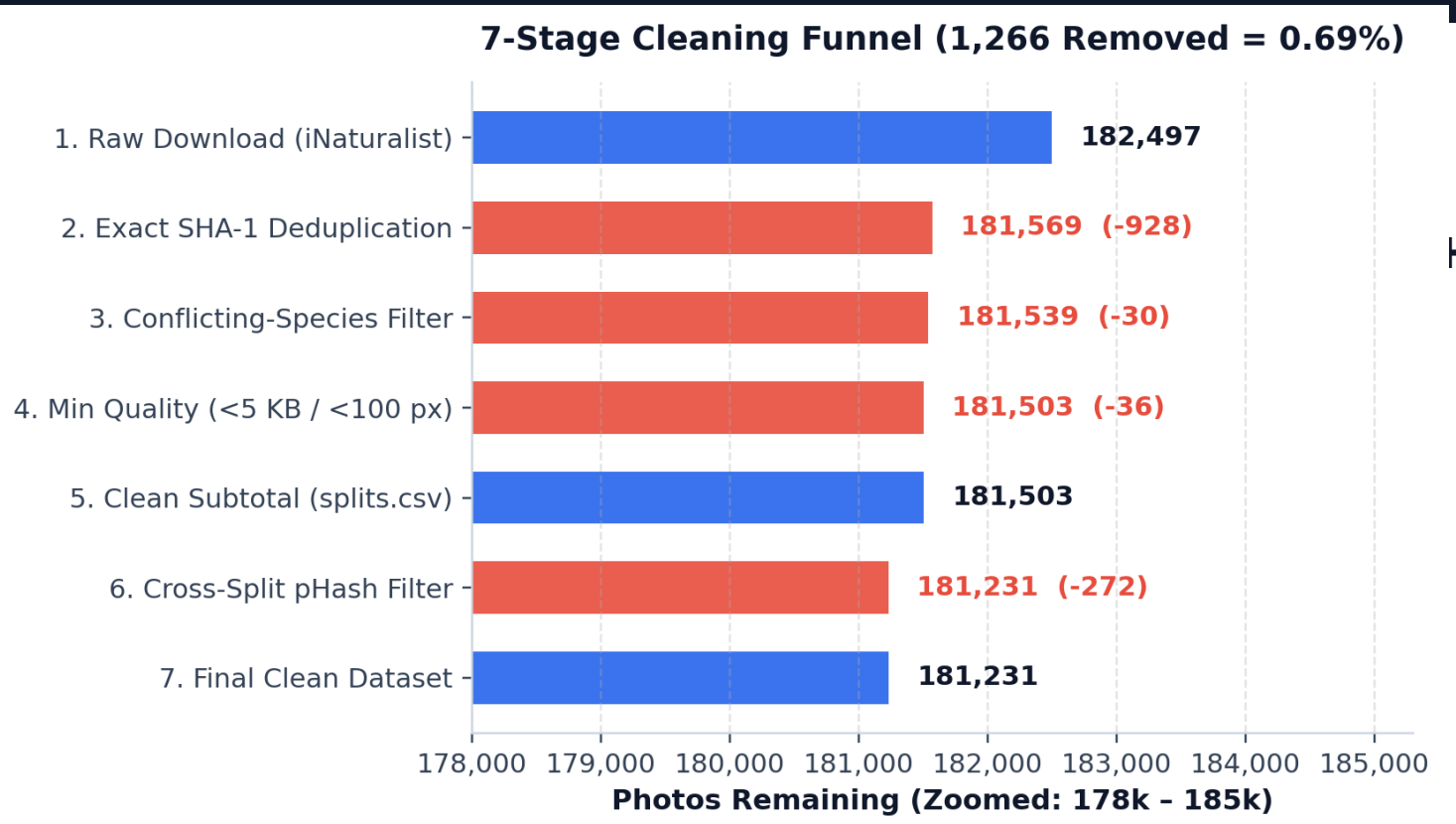
DATASET SCALE	TAXONOMIC SCOPE	MEDICAL PRIORITY	SPLIT INTEGRITY
181,231	**208 Classes**	**18 WHO Species**	**0 Leakage**
VERIFIED SNAKE PHOTOS	**FROM 289 INDIAN SPECIES**	**22,993 HIGH-RISK PHOTOS**	**OBSERVER-GROUPED SPLITS**
95,877 observations by 25,070 photographers	Spanning 14 families and 77 genera	6 Cat 1 Critical & 12 Cat 2 Important	StratifiedGroupKFold on `observer_id`

Milestone 2 Scope & Presentation Agenda

I. DATASET ENGINEERING	II. TAXONOMIC & CLINICAL SAFETY ANALYSIS	III. SPLIT HYGIENE, QUALITY & MODELING
• Data Sourcing & 7-Stage Cleaning Funnel • 31-Column Schema & 5-Partition Design	• Species Long-Tail Distribution • 14 Families & 77 Genera Clade Structure • WHO Category 1 & 2 Venomous Coverage • 3-Tier Strategy & Milestone-1 Core-27 Link	• Photographer Isolation & Post-2022 Test Set • Image Resolution & Aspect Ratio Audit • India Hotspots & Monsoon Seasonality • Milestone 3 Modeling Architecture Plan

Group 5: Jayesh Bansal (Team Lead) · Vikas · Puneet Bajaj · Manas Rastogi

From 182,497 Raw Downloads to 181,231 Clean, Deduplicated Photos

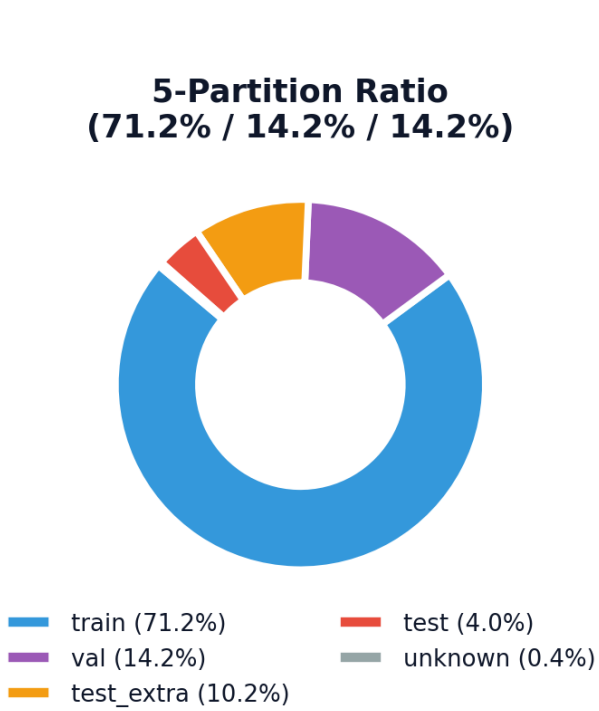


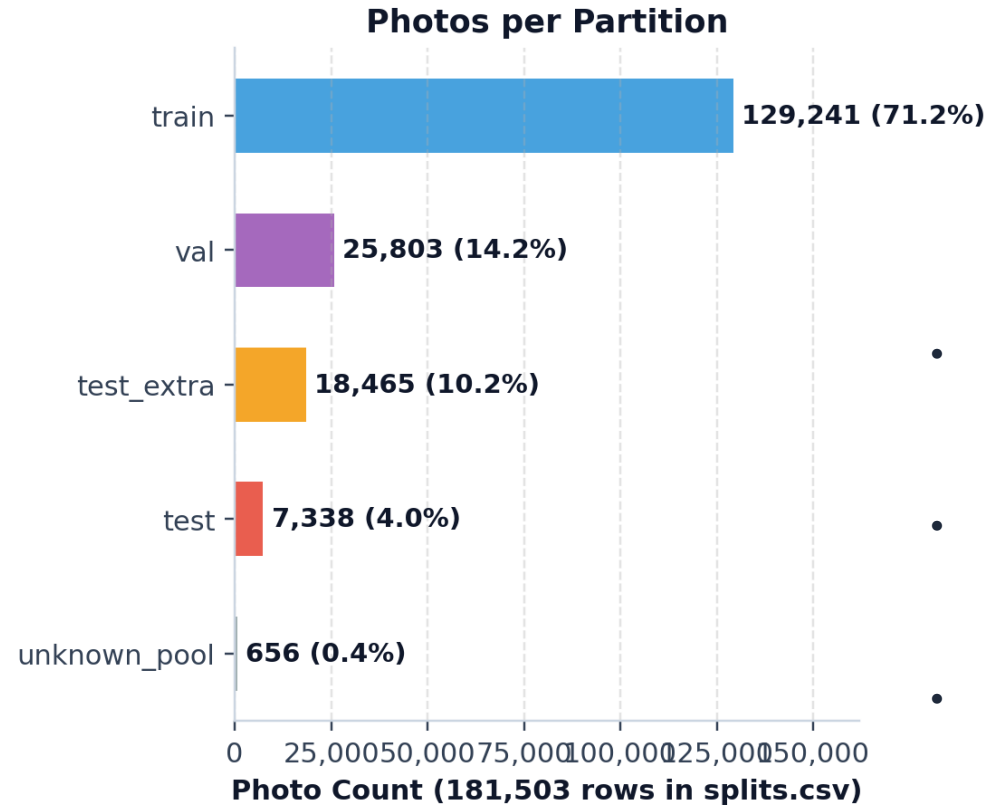
How We Cleaned the Raw Data

- **Primary Source (iNaturalist India)**: Selected for research-grade taxonomy, observer IDs, and GPS. SnakeCLEF 2023 (4,989 photos) was skipped because 100% are iNaturalist duplicates.
- **File & Label Hygiene (-994 logged in `excluded.csv`)**: Removed **928** exact SHA-1 duplicates, **30** conflicting-species labels, and **36** broken/tiny images (<5 KB or <100 px) → **181,503** rows in `splits.csv`.
- **Perceptual Hash Filter (-272)**: Removed cross-split pHash near-duplicates → **181,231** final images.

ARTEFACT FILE	ROW / RECORD COUNT	ROLE IN REPRODUCIBLE KAGGLE DATASET
`splits.csv` & `classes.csv`	181,503 photo rows · 208 class rows	Master photo manifest (31 columns) and active class lookup table
`excluded.csv` & `split_report.json`	994 excluded rows · 1 audit JSON	Full audit trail of rejected photos and automated split verification metrics

99.18% Complete Metadata & Balanced 71.2% / 14.2% / 14.2% Split



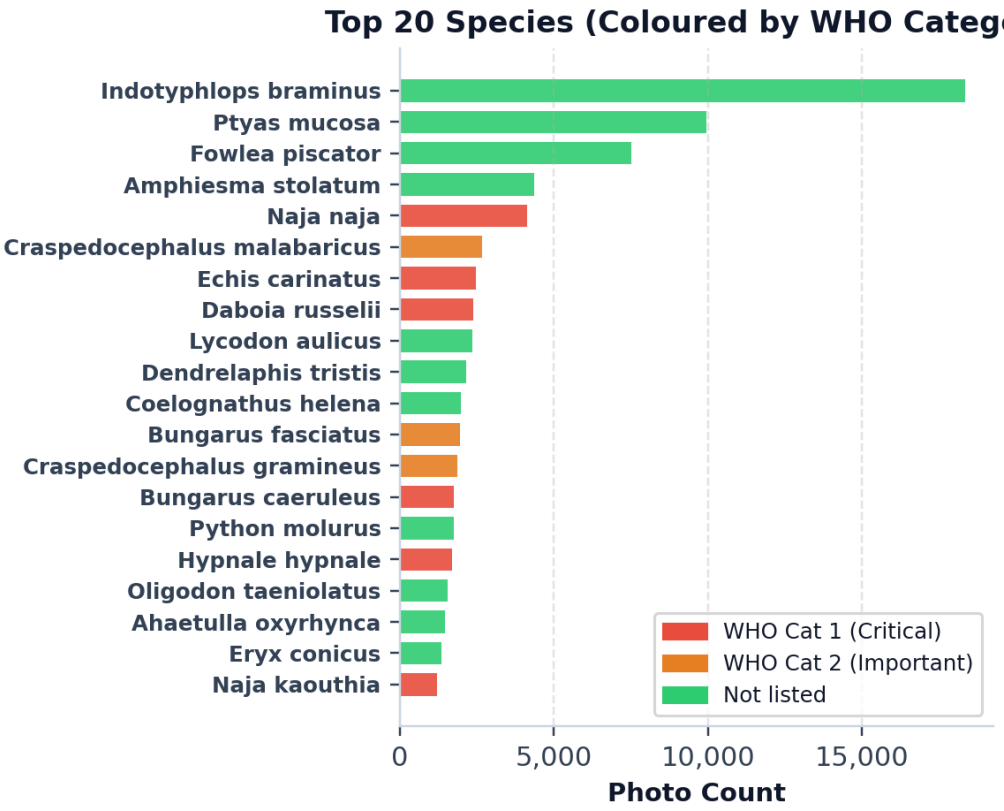


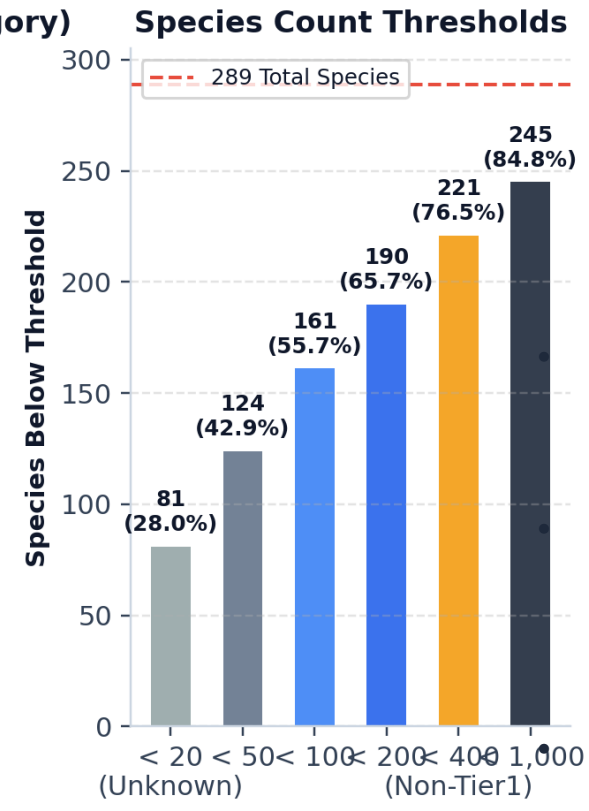
ema & Partition Design

- **o Missing Values in 29 of 31 Columns**: Complete IDs, onomy, dimensions, SHA-1 hashes, and safety flags oss all 181,503 rows.
- **18% GPS Coverage**: Only `latitude` and `longitude` e 0.82% missing (1,495 obscured rows; 180,008 d).
- **anced 70 / 15 / 15 Architecture**: `train` (71.2%), `val` .2%), and combined `test` (4.0%) + `test_extra` .2%) = **14.2%**, plus unknown_pool (0.4%).

COLUMNS)	KEY COLUMNS IN SPLITS.CSV	COMPLETENESS
Identifiers & Provenance	`file`, `obs_id`, `photo_id`, `observer_id`, `sha1`, `license`, `cc_licensed`	**100.0%** (0 missing)
Taxonomy & Safety Flags	`species`, `genus`, `family`, `class_id`, `who_category`, `tier`, `m1_core27`	**100.0%** (0 missing)
Spatio-Temporal & Specs	`observed_on`, `width`, `height`, `bytes`, `in_india` (100%) · `latitude`, `longitude` (99.18%)	**99.18%** (1,495 missing GPS)

Power-Law Long Tail: Median 79 Photos/Species; 55.7% Have <100 Photos





'eme Class Imbalance

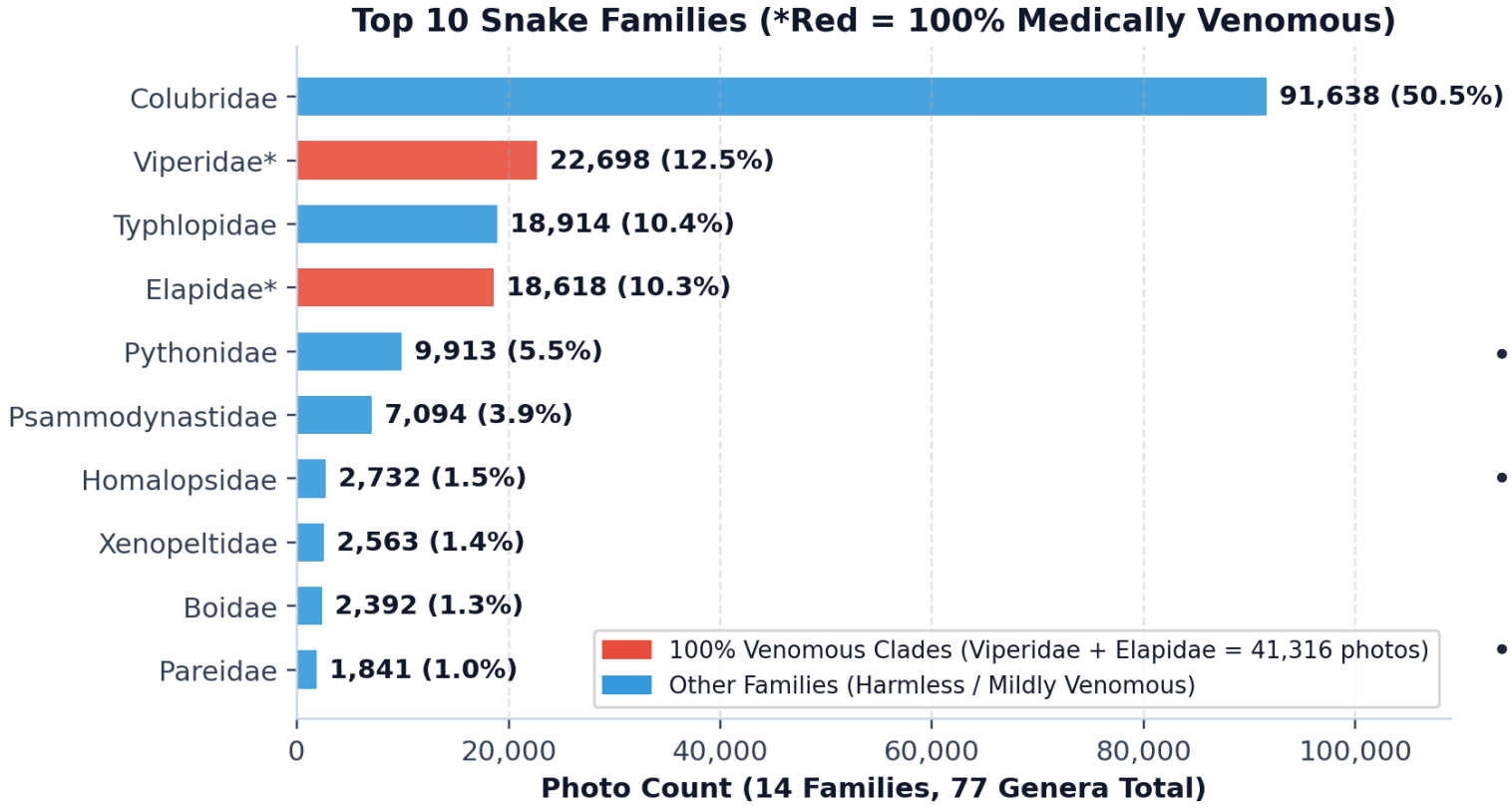
,340 : 1 Dynamic Range: Across 289 species, counts ıge from **min = 1** to **max = 18,340** (*Indotyphlops ıminus*), with a **median of 79 photos**.

ad Classes: Top 4 non-venomous species (*I. ıminus* 18,340, *Ptyas mucosa* 9,943, *Fowlea piscator* 24, *Amphiesma stolatum* 4,351) account for 22.1% of photos.

ıy Standard Loss Fails: 161 species (55.7%) have 0 photos and 221 (76.5%) have <400, requiring tiered focal loss.

THRESHOLD CUTOFF	< 20 PHOTOS	< 50 PHOTOS	< 100 PHOTOS	< 200 PHOTOS	< 400 PHOTOS	< 1,000 PHOTOS
Species Below Cutoff (of 289)	81 (28.0%)	124 (42.9%)	161 (55.7%)	190 (65.7%)	221 (76.5%)	245 (84.8%)

Colubridae Leads Volume (50.5%); Viperidae & Elapidae Add 41,316 Venomous Photos



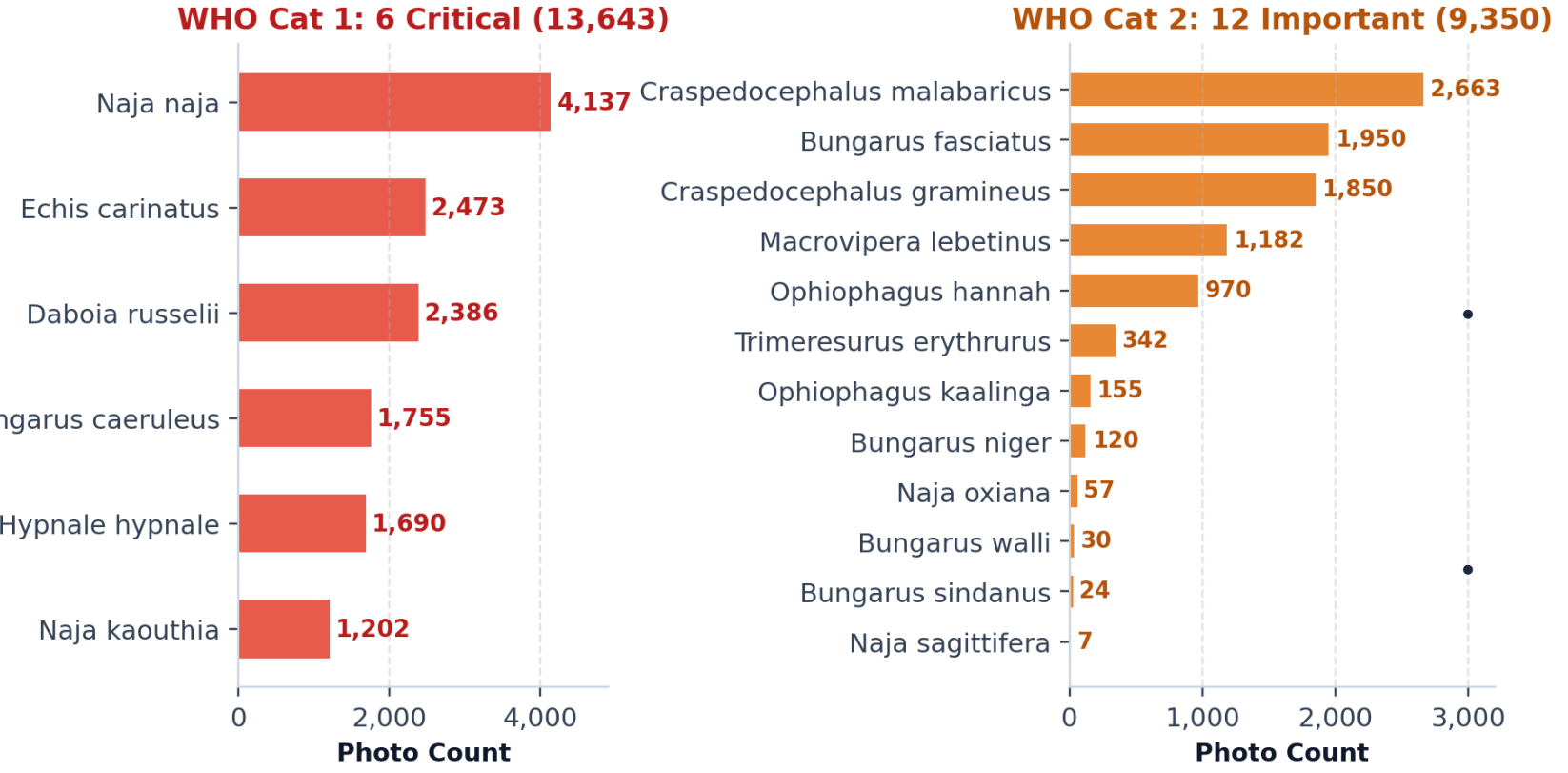
ade-Level Insights

- **14 Families & 77 Genera**: Provides clean biological structure for hierarchical multi-task supervision.
- **Colubridae Dominates (91,638 photos, 50.5%)**: Mostly harmless or mildly venomous snakes (rat snakes, keelbacks, vine snakes, wolf snakes).
- **100% Venomous Families (22.8% of Dataset)**: **Viperidae** (22,698 photos, 12.5%) and **Elapidae** (18,618 photos, 10.3%) are entirely medically venomous — family-level prediction alone flags severe envenomation risk.

Viperidae & Elapidae	41,316 (22.8%)	**100% Venomous**: Vipers/pit vipers (*Daboia, Echis, Hypnale, Craspedocephalus*) & Elapids (*Naja, Bungarus, Ophiophagus*)
Colubridae & Typhlopidae	110,552 (60.9%)	**Harmless / Rear-Fanged**: Rat snakes (*Ptyas*), keelbacks (*Fowlea*), blind snakes (*Indotyphlops*)

Taxonomic Hierarchy Across 14 Families & 77 Genera

22,993 Photos (12.7%) Cover 18 Medically Important WHO Category 1 & 2 Species

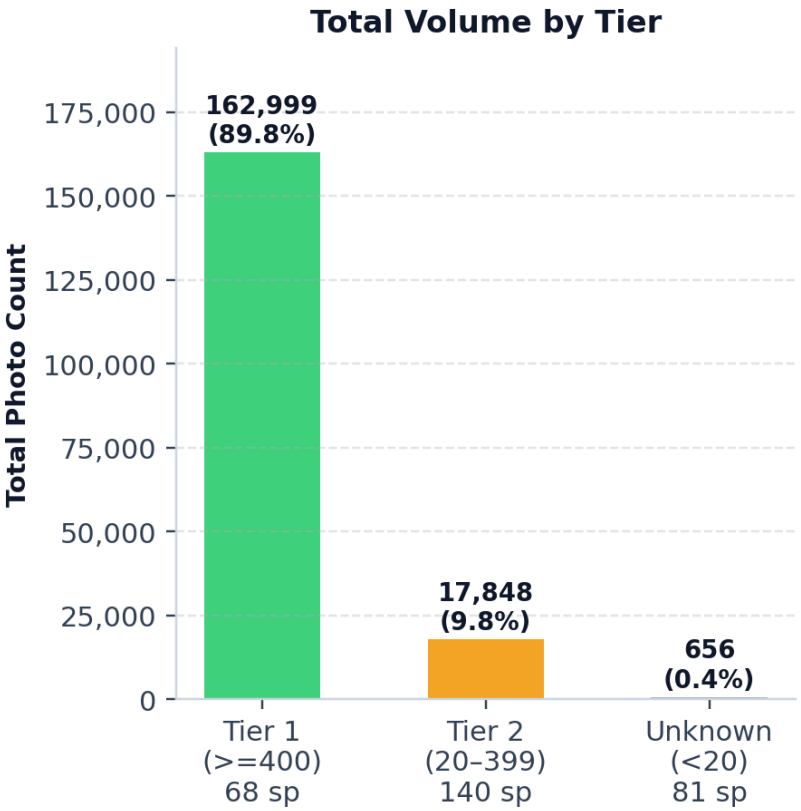


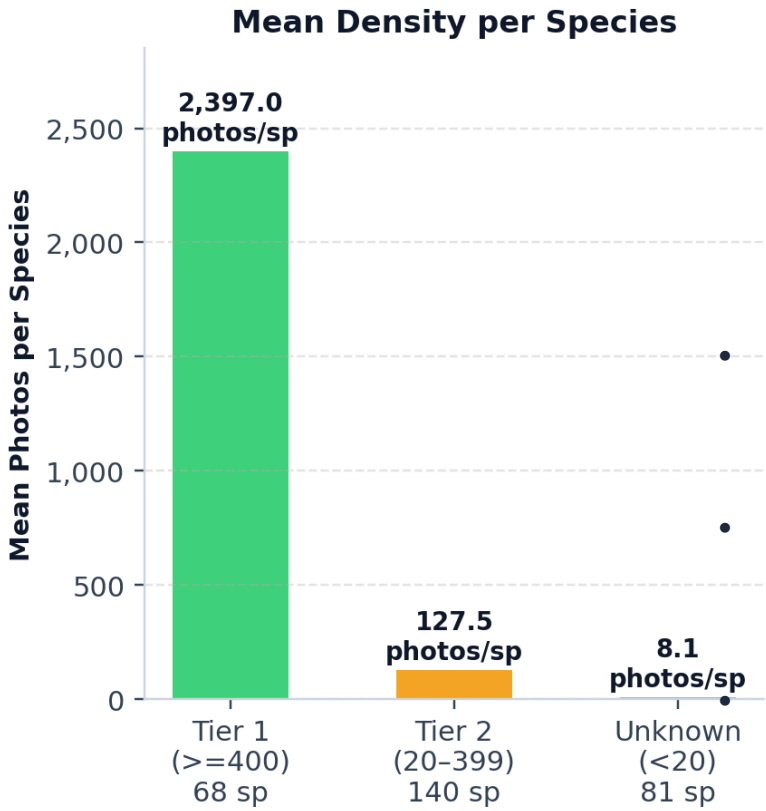
al Risk Breakdown

- **Category 1 (6 Species — 13,643 Photos, 7.5%)**: ...s >95% of Indian snakebite deaths. Every species ...1,200 photos (*Naja naja* 4,137, *Echis carinatus* ..., *Daboia russelii* 2,386, *Bungarus caeruleus* 1,755, ...*ale hypnale* 1,690, *Naja kaouthia* 1,202).
- **Category 2 (12 Species — 9,350 Photos, 5.2%)**: ...idary regional venomous snakes led by *C.* ...*aricus* (2,663), *B. fasciatus* (1,950), *C. gramineus* ...), *M. lebetinus* (1,182), and *O. hannah* (970).
- **...ested (271 Species — 158,510 Photos, 87.3%)**: Confident non-venomous ID prevents panic and antivenom misuse.

WHO RISK LEVEL	SPECIES COUNT	PHOTO VOLUME	MODELING PRIORITY IN SERPENTGUARD AI
Category 1 (Critical)	6 species	13,643 (7.5%)	Non-negotiable high recall via asymmetric false-negative penalty
Category 2 (Important)	12 species	9,350 (5.2%)	5 species in Tier-1, 6 in Tier-2, 1 (*N. sagittifera*, 7 photos) in `unknown_pool`

Tier-1 (68 Species) Holds 89.8% of Photos; All 27 M1 Species Are in Tier-1



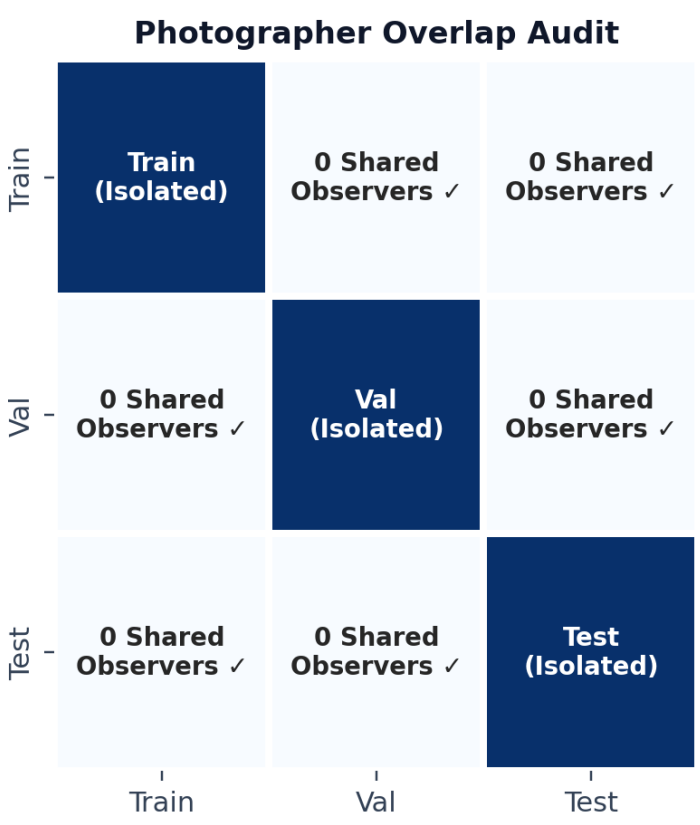


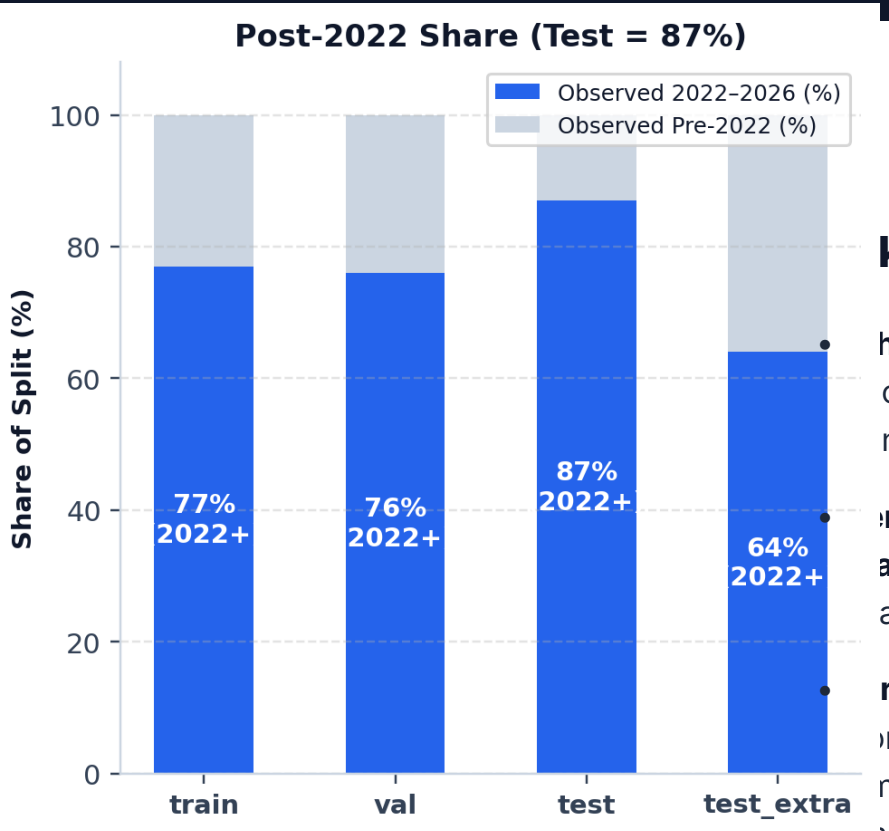
v Tiers Solve the Long Tail

- **er1 (≥400 photos — 68 Species)**: **162,999 photos 9.8%)**, averaging **2,397.0 photos/species**. Forms our gh-accuracy species backbone.
- **er2 (20–399 photos — 140 Species)**: **17,848 photos .8%)**, averaging **127.5 photos/species**. Trained with ss-balanced focal loss and augmentation.
- **known_pool (<20 photos — 81 Species)**: **656 photos .4%)**, averaging **8.1 photos/species**. Held out to nchmark open-set abstention.

MILESTONE-1 BASELINE CONTINUITY	SPECIES COUNT	PHOTO RANGE IN M2	TOTAL PHOTOS
M1 Core-27 Species (`m1_core27 == True`)	All 27 in `tier1` (5 WHO Cat 1, 2 WHO Cat 2, 20 Not Listed)	682 (*Ahaetulla farnsworthi*) to 18,340 (*I. braminus*)	**77,130 photos**

Zero Photographer Overlap Across Splits & 87% Post-2022 Test Set





kage-Free Evaluation

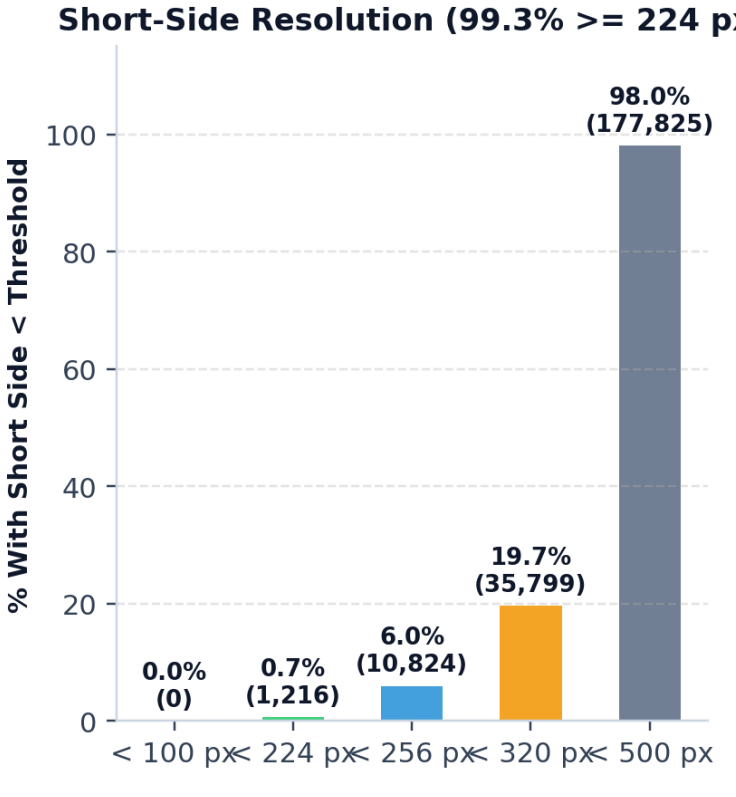
hy Group by `observer_id`? Observers upload multi-oto bursts of the same snake (~1.89 photos/obs). ndom splitting leaks identical snakes into test.

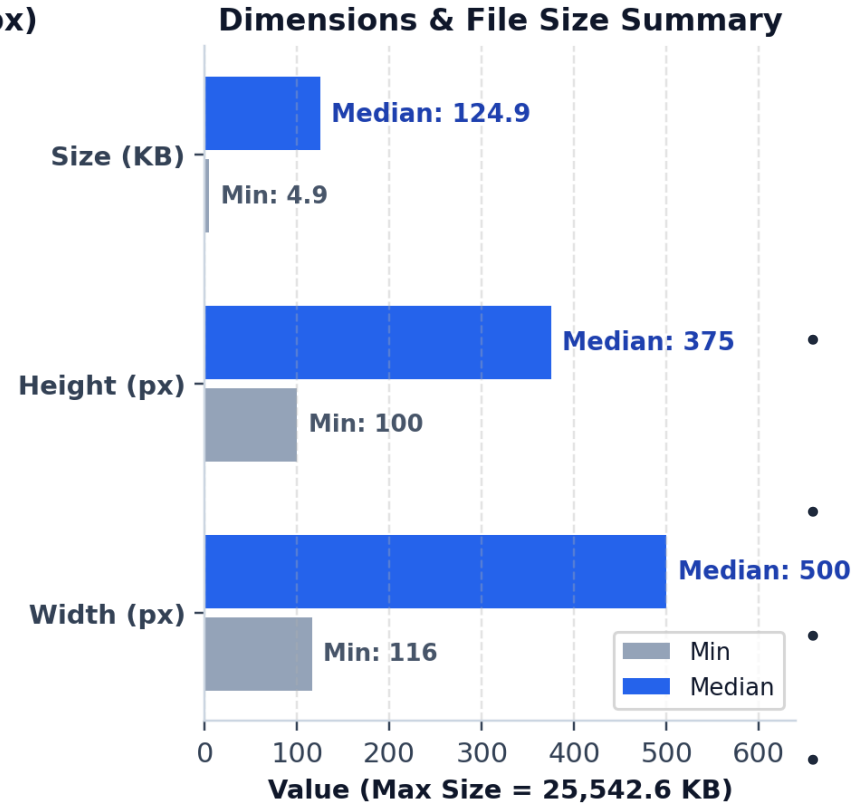
rified Zero Overlap: Using `StratifiedGroupKFold`, **ain ∩ Val = 0**, **Train ∩ Test = 0**, and **Val ∩ Test = 0** ared observers.

rward Temporal Cutoff: **87% of `test` photos** come m 2022–2026 (vs. 77% in `train` and 76% in `val`), nulating real-world deployment on modern artphone cameras.

SPLIT NAME	YEAR SPAN	% OBSERVED 2022–2026	OBSERVER OVERLAP CHECK
`train` (129,241) · **`val`** (25,803)	1968–2026 · 1980–2026	77% · 76%	**0 shared observers ✓**
`test` (7,338) · **`test_extra`** (18,465)	1970–2026 · 1990–2026	**87%** · 64%	**0 shared observers with Train/Val ✓**

99.3% of Photos Meet ≥224 px Input Size (Median 500×375 px, 124.9 KB)



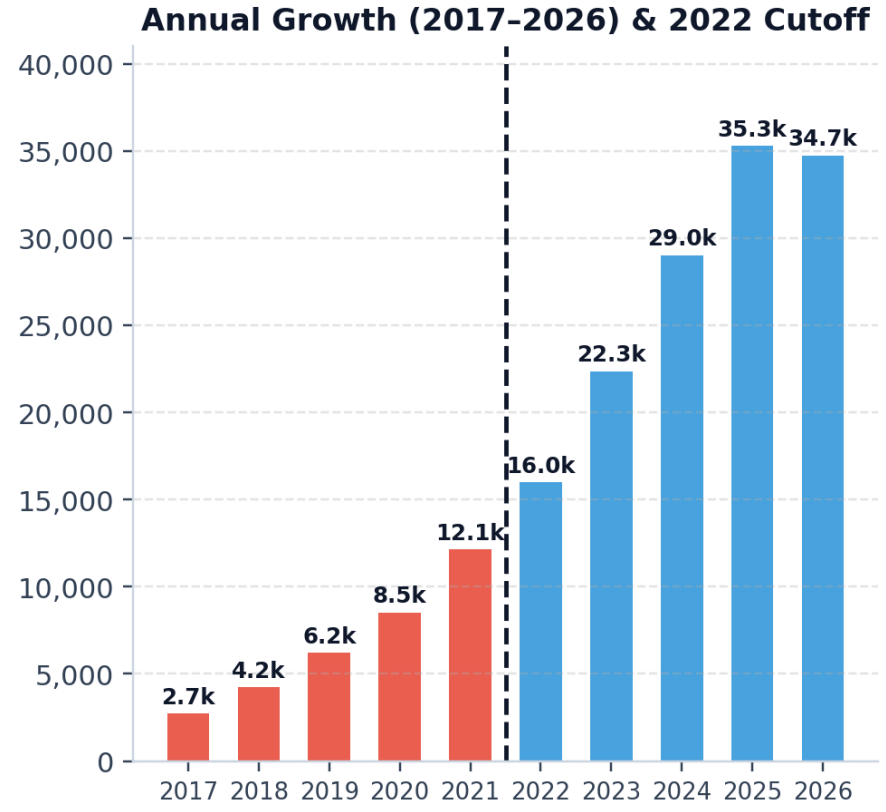


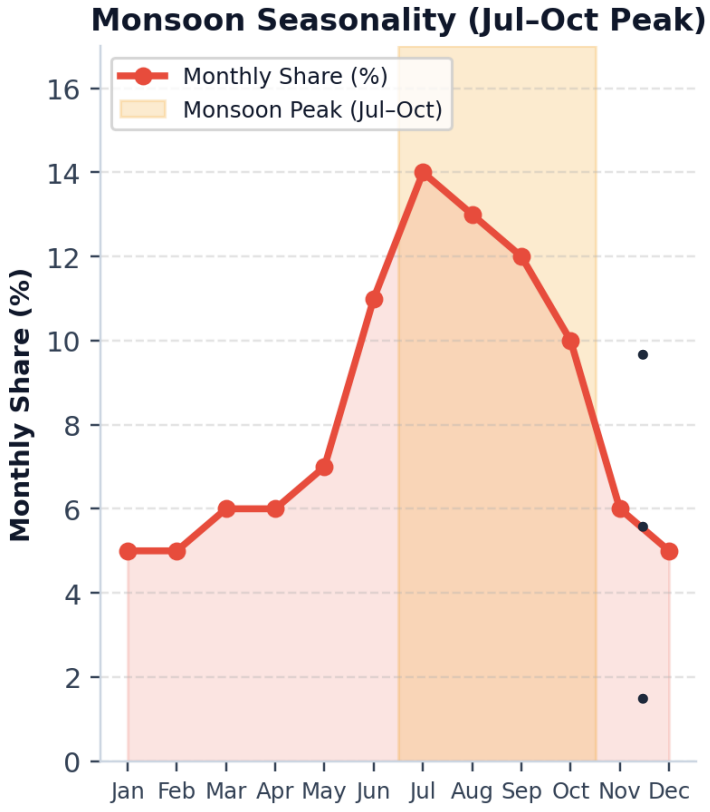
ge Specifications

- **nsistent 500 px Medium Standard**: Width ranges **6–500 px (median 500 px)**; Height ranges **100–500 (median 375 px)**.
- **mpact File Footprint**: Ranges **4.9 KB to 25,542.6 KB** th a **median of 124.9 KB**.
- **pect Ratio Peaks**: Clusters at **1.33 (4:3 landscape)**, **75 (3:4 portrait)**, and **1.0 (square)**.
- **ady for 224/384px Backbones**: Only **0.7% (1,216 otos)** have a short side <224 px, preventing upscaling blur.

SHORT-SIDE RESOLUTION CUTOFF	< 100 PX	< 224 PX (VIT/CNN)	< 256 PX	< 320 PX	< 500 PX (INAT CAP)
Photos Below Cutoff	0 (0.0%)	**1,216 (0.7%)**	10,824 (6.0%)	35,799 (19.7%)	177,825 (98.0%)

Western Ghats & Gangetic Hotspots; Sightings Surge During Jul–Oct Monsoon





ial & Temporal Patterns

tial Distribution: **180,008 photos (99.2%)** have ; **75,306 (41.5%)** fall inside the India bounding box 38° N, 68°–98° E).

ional Hotspots: Dense clusters in the **Western ts** (8°–15° N) and **Gangetic plains**; Northeast India is ser due to lower observer density.

Post-2017 Growth & Monsoon Peak: Volume bed from 2,713 (2017) to **35,283 (2025)** and **34,749 26)**, peaking every **July–October monsoon**.

Y FOR SERPENTGUARD AI

India Bounding Box vs. Worldwide	75,306 (41.5%) India · 104,702 (57.7%) Global	Global photos boost rare Indian species in training; primary benchmark stays India-focused
Monsoon Surge (Jun–Oct)	~60% of annual sightings	Coincides directly with peak agricultural activity and hospital snakebite admissions

MILESTONE 3 MODELING ARCHITECTURE & PROJECT TEAM

EDA-Driven Modeling Decisions for Milestone 3 & Group 5 Team

Processed Dataset on Kaggle: `puneetbajajiitm/serpentguard-m2-splits` · IIT Madras BS in Data Science & AI Lab (2026 T3)

CORE EDA FINDING	QUANTIFIED EVIDENCE	MILESTONE 3 ENGINEERING ACTION
Severe Long-Tail Imbalance	Median 79 photos/sp; 55.7% of species have <100 photos	**Class-Balanced Focal Loss + Augmentation**: Separate Tier-1 (68 sp) & Tier-2 (140 sp) F1 tracking
High-Stakes WHO Cat 1 & 2	6 Cat 1 (13,643 photos) & 12 Cat 2 (9,350 photos)	**Asymmetric Safety Loss**: Penalize venomous false negatives heavily + dedicated WHO risk head
81 Ultra-Rare Species (<20)	656 photos held out in `unknown_pool`	**Open-Set Abstention**: Calibrate energy/entropy threshold to output safe "Unknown Snake" warning
Strong Clade Structure	14 families, 77 genera; Viperidae + Elapidae = 100% venomous	**Hierarchical Multi-Task Heads**: Jointly predict Species (208) + Genus (77) + Family (14) + WHO (3)
500×375 px Resolution & Splits	99.3% ≥224 px; 0 observer overlap; 87% of Test is 2022+	**224/384px Backbone + SAM 2.1 Masks**: Train on `train`, tune on `val`, benchmark on 2022+ `test`

Group 5 Project Team — IIT Madras DS & AI Lab (2026 T3)

JAYESH BANSAL · TEAM LEAD	VIKAS · TEAM MEMBER	PUNEET BAJAJ · TEAM MEMBER	MANAS RASTOGI · TEAM MEMBER
Project Architecture, Problem Formulation & Safety Evaluation	Taxonomic & WHO Clinical Risk Analysis, Literature Review	iNaturalist Data Engineering, Deduplication & Kaggle Splits	Exploratory Data Analysis, Quality Audit & Baseline Pipeline